Interspeech 2026
27 September - 1 October 2026, Sydney, Australia

Schrödinger Bridge Mamba for One-Step Speech Enhancement

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Abstract

We present Schrödinger Bridge Mamba (SBM), a novel model for efficient speech enhancement by integrating the Schrödinger Bridge (SB) training paradigm and the Mamba architecture. Experiments of joint denoising and dereverberation tasks demonstrate SBM outperforms strong generative and discriminative methods on multiple metrics with only one step of inference while achieving a competitive real-time factor for streaming feasibility. Ablation studies reveal that the SB paradigm consistently yields improved performance across diverse architectures over conventional mapping. Furthermore, Mamba exhibits a stronger performance under the SB paradigm compared to Multi-Head Self-Attention (MHSA) and Long Short-Term Memory (LSTM) backbones. These findings highlight the synergy between the Mamba architecture and the SB trajectory-based training, providing a high-quality solution for real-world speech enhancement. Demo page: `https://sbmse.github.io`

Index Terms: Schrödinger Bridge, Mamba, state-space models, trajectory modeling, speech enhancement

1. Introduction

Deep generative models are increasingly employed for speech enhancement (SE). By learning the clean speech distribution directly, generative models achieve superior perceptual quality and reconstruct fine details lost in deterministic regression [1]. While earlier score-based generative models showed promising SE performance, they suffer from the *mean prior mismatch* issue [1]. To resolve this, the Schrödinger Bridge (SB) paradigm models the optimal transport (OT) path via stochastic differential equations (SDEs), seamlessly transporting the degraded prior directly to the clean target [2, 3, 4, 5, 6]. SB methods have demonstrated remarkable success across SE tasks (denoising/dereverberation [2, 3, 4, 5, 6], super-resolution [7], inpainting [8]), image generation [9] and text-to-speech [10]. Following broader successes [11, 12], most SB-based SE methods adopt the NCSN++ architecture as their backbone (SB-NCSN++) [2, 4, 5, 6].

However, the slow inference of these SB-based methods (often requiring > 10 iterative steps) hinders real-time application. To accelerate generation, recent approaches explored specialized schedulers and solvers [7], adversarial training [5, 13] or consistency trajectory modeling [6, 14]. While some achieve one-step inference in denoising or dereverberation tasks [5, 6], existing works typically overlook the inherent synergy between the SB paradigm and the backbone architecture, leaving significant room for improvement in efficient modeling.

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Recently, the selective state-space model Mamba [15] has emerged as a powerful architecture for modeling long-range audio dependencies. Although adopted in SE tasks (e.g., oSpatialNet [16], SEMamba [17], USEMamba [18]), these works utilize deterministic mapping or masking training strategy, failing to exploit the potential of generative trajectory learning.

In this work, we propose ***Schrödinger Bridge Mamba (SBM)***, the first framework to synergize the Schrödinger Bridge (SB) paradigm with the Mamba architecture for one-step speech enhancement leveraging generative trajectory guidance. Our core argument is that aligning *training paradigm* with the *backbone architecture*'s inductive bias is essential for efficiency and effectiveness. Unlike 'blind' deterministic mapping or masking that only sees the start and end points, the SB paradigm explicitly calculates intermediate states x_t along the optimal transport path bridging the degraded and clean speech distributions [2]. These states x_t are utilized as training inputs, effectively acting as 'anchors' to guide the backbone model's learning of the underlying evolution process. The Mamba architecture is well-suited for the SB training paradigm: (1) As a state-space model, Mamba naturally emulates the process of state evolution [15]; (2) Its selective mechanism facilitates adaptive context modeling, allowing the network to learn the dynamics of the optimal transport path [15, 19]. Overall, SBM 'distills' the SB transformation into the Mamba dynamics during the training process, enabling high-fidelity reconstruction in a single inference step.

We evaluate SBM on joint denoising and dereverberation SE tasks that represent common real-life scenarios. Experiments on DNS and VoiceBank-Demand test sets demonstrate that SBM outperforms conventional SB-NCSN++ models, mapping-trained Mamba model, flow-matching variant, one-step SB variants and strong discriminative model (ZipEnhancer [20]) in real-world and reverberant scenarios, and achieves the best real-time factor (RTF) among these methods. Furthermore, ablation studies replacing Mamba with Multi-Head Self-Attention (MHSA) and Long Short-Term Memory (LSTM) reveal that the SB paradigm consistently outperforms the mapping paradigm across architectures, highlighting the potential of the training paradigm. Moreover, the superior performance of Mamba in these experiments highlights its effectiveness in capturing trajectory dynamics. Collectively, these results justify the SBM design and underscore the potential of integrating continuous-time diffusion processes with state-space backbones, offering valuable insights for the broader evolution of continuous-time sequence modeling in complex audio tasks.

2. Schrödinger Bridge Mamba

Schrödinger Bridge Mamba (SBM) leverages the generative optimal transport (OT) trajectory of Schrödinger Bridge (SB)

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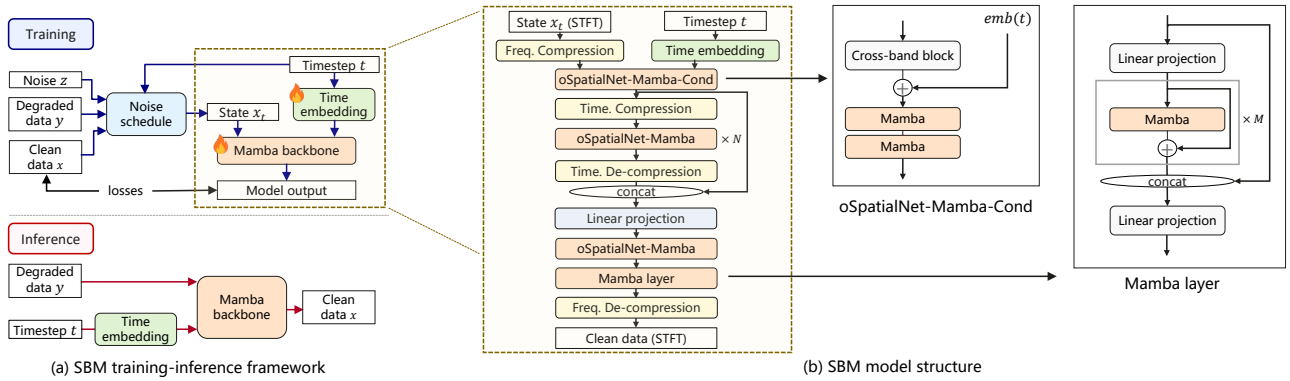


Figure 1: *Overview of the SBM training-inference framework and the Mamba-based model architecture.*

to train a selective state-space Mamba backbone. This section details the SB formulation, the model architecture and the training-inference framework.

2.1. Schrödinger Bridge Formulation

Unlike standard diffusion models that rely on Gaussian priors leading to *mean prior mismatch* issue [1], SBM formulates speech enhancement as an OT process directly between the degraded speech distribution p_T and the clean speech distribution p_0. Mathematically, this process is governed by a system of SDEs, and the intermediate states $\{x_t\}_{0:T}$ along the SB stochastic process can be constructed using paired data samples (clean speech $x \sim p_0$ and degraded speech $y \sim p_T$) [2, 21, 22]. To enable efficient training for speech enhancement, following the tractability of SB OT formulation [2, 4, 5, 6], the intermediate states x_t at timesteps $t \in [0, 1]$ can be explicitly parameterized as an interpolation of the boundary conditions plus a stochastic Wiener process term. Specifically, x_t is sampled via

$$x_t = \mu_x(t) + \sigma_x(t)z, \quad z \sim \mathcal{N}_C(0, \mathbf{I}) \tag{1}$$

where $\mu_x(t) = w_x(t)x + w_y(t)y$. The variance $\sigma_x(t)$ and coefficients $w_x(t)$, $w_y(t)$ are derived from the Variance Exploding (VE) noise schedule parameters [2] in our implementation. Crucially, Eq.(1) serves as the data generator during training: For each batch, we sample random timesteps t and compute states x_t, training the model to reconstruct the clean target x by learning the underlying data distributions.

2.2. Mamba-based Architecture

The architectural design of SBM is motivated by the structural resemblance between the SB theory and Mamba's discretized recurrence $h_t = \mathbf{A}h_{t-1} + \mathbf{B}u_t, y_t = \mathbf{C}h_t$ [15]. Viewed as a controlled system, $\mathbf{A}h_{t-1}$ represents natural, uncontrolled dynamics [23], while the input-dependent parameters $(\mathbf{B}, \mathbf{C})$ act as control terms. Training Mamba thus resembles learning the optimal control strategy inherent to SB [22], where the selective mechanism dynamically parameterizes the transport path based on current states. Furthermore, recently proven mathematical optimality of Mamba in capturing Markovian sequence dynamics [24] also suggests its suitability for modeling the stochastic evolution process of the SB paradigm [22].

Guided by these insights, we construct a Mamba-based model as shown in Figure 1(b). Following the common practice in related work [2, 4, 5, 6, 16, 17, 18], we represent audio using STFT spectra. The main component *oSpatialNet-Mamba* is implemented following the design in [16]. To incorporate the timesteps defined by the SB training paradigm, we train a *Gaussian Fourier* module to embed timesteps [2, 4], and the time embedding *emb(t)* is added to the input of Mamba in the *oSpatialNet-Mamba* module, which is termed *oSpatialNet-Mamba-Cond* in SBM. Inspired by [16, 25], to address the temporal limitations of oSpatialNet, we integrate a fullband *Mamba layer* to capture global spectral dynamics and inter-frame dependencies. For streaming feasibility, the Mamba backbone operates with a small lookahead of 2-4 frames, yielding an algorithmic latency under 40 ms to balance causality and enhancement quality.

2.3. Training and One-Step Inference

Figure 1(a) sketches the training and inference process of SBM. The Mamba backbone takes noisy intermediate states x_t and timestep embeddings as inputs to learn the distribution of clean data x. Typical denoising objectives as in diffusion models can be applied [5], among which the data prediction loss has shown good performance for speech enhancement [2, 4, 5, 6]. We employ a comprehensive data prediction loss combining magnitude and complex domain constraints: $L = \lambda_1 L_{mse}(S, \hat{S}) + \lambda_2 L_{mse}(|S|, |\hat{S}|) + \lambda_3 L_{mr,mse}(S, \hat{S}) + \lambda_4 L_{mr,mse}(|S|, |\hat{S}|)$, where S and $\hat{S}$ are the target and modeled STFT, mr refers to multi-resolution and $|S|$ refers to magnitude spectrum.

While standard SB methods require iterative solving of the reverse SDE, SBM is designed for one-step generation. During inference, we set the timestep to the start of the reverse process ($t = 1$, corresponding to the degraded prior). The model directly reconstructs the clean target in a single forward pass. This drastically reduces latency while leveraging the distributional modeling capability learned from the SB trajectory.

3. Experiments and Results

3.1. Experimental Setup

To cover common real-life scenarios, we focus on joint denoising and dereverberation tasks for speech enhancement. Audio clips are arbitrarily segmented into 3-second segments on-the-fly during training. For the STFT representation, we use a NFFT of 512, a Hann window of length 480 and a hop length of 160. The SBM model is trained using AdamW optimizer with a cosine learning rate scheduler starting from $lr = 1e-3$. The hyper-parameters of the loss $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are set to 1000, 1000, 500, 500.

Table 1: *Performance comparison on benchmark test sets DNS With Reverb, DNS No Reverb and VoiceBank-Demand. "RTF" stands for real-time factor, measured on a GPU by averaging over 10 pieces of 10s audio samples.*

Test set	Model	Para.[M]	RTF↓	SIG↑	BAK↑	OVRL↑	P808MOS↑	NISQA↑	SpeechBERTScore↑	Similarity↑	PESQ↑	ESTOI↑
DNS With Reverb	SB-NCSN++(50)	25.16	0.767	3.395	3.938	3.085	3.884	3.879	0.703	0.947	1.532	0.629
	SB-NCSN++(10)	25.16	0.156	3.425	3.97	3.126	3.896	**4.087**	0.728	0.947	1.597	0.671
	SB-NCSN++(1)	25.16	0.0155	3.281	3.871	2.948	3.559	3.323	0.730	0.943	1.586	0.67
	SBCTM	65.98	0.02	2.927	3.859	2.623	3.035	2.578	0.494	0.859	1.218	0.281
	SB-UFOGen	25.16	0.0152	2.964	3.664	2.596	3.337	2.131	0.662	0.918	1.383	0.601
	ZipEnhancer	2.04	0.105	2.771	3.164	2.264	2.85	1.926	0.534	0.84	1.227	0.288
	Mamba-base	3.61	**0.0048**	3.202	3.898	2.895	3.665	3.514	0.766	0.963	1.895	0.696
	FM-Mamba	3.93	**0.0048**	3.164	3.956	2.887	3.461	3.148	0.682	0.930	1.643	0.634
	SBM	3.93	**0.0048**	**3.653**	**3.995**	**3.367**	**4.055**	4.053	**0.784**	**0.965**	**1.971**	**0.71**
DNS No Reverb	SB-NCSN++(50)	25.16	0.767	3.515	4.02	3.231	4.01	4.233	0.78	0.976	1.656	0.8
	SB-NCSN++(10)	25.16	0.156	3.477	4.069	3.225	3.968	4.472	0.792	0.974	1.651	0.824
	SB-NCSN++(1)	25.16	0.0155	3.387	4.087	3.148	3.665	3.892	0.787	0.962	1.669	0.791
	SBCTM	65.98	0.02	3.523	4.128	3.298	3.906	4.574	0.870	0.986	2.835	0.898
	SB-UFOGen	25.16	0.0152	3.443	4.007	3.161	3.76	3.719	0.793	0.964	1.754	0.803
	ZipEnhancer	2.04	0.105	3.578	**4.136**	**3.35**	4.015	4.289	**0.922**	**0.992**	**3.419**	**0.952**
	Mamba-base	3.61	**0.0048**	3.515	4.053	3.253	3.997	4.468	0.887	0.99	2.73	0.916
	FM-Mamba	3.93	**0.0048**	3.446	4.057	3.192	3.783	4.088	0.802	0.968	1.964	0.843
	SBM	3.93	**0.0048**	**3.751**	4.129	3.292	**4.187**	**4.657**	0.893	0.99	2.825	0.921
VoiceBank-Demand	SB-NCSN++(50)	25.16	0.767	3.415	4.005	3.134	3.55	4.237	0.764	0.942	1.887	0.753
	SB-NCSN++(10)	25.16	0.156	3.382	4.05	3.131	3.61	4.688	0.795	0.953	2.112	0.79
	SB-NCSN++(1)	25.16	0.0155	3.337	4.087	3.106	3.394	4.41	0.778	0.934	2.165	0.76
	SBCTM	65.98	0.02	3.4	4.085	3.156	3.533	4.646	0.891	0.978	3.558	0.868
	SB-UFOGen	25.16	0.0152	3.384	4.06	3.133	3.371	4.132	0.768	0.926	2.152	0.758
	ZipEnhancer	2.04	0.105	**3.484**	3.988	3.183	3.544	4.621	**0.925**	**0.986**	**3.628**	**0.894**
	Mamba-base	3.61	**0.0048**	3.294	3.904	2.969	3.498	4.441	0.814	0.975	2.612	0.831
	FM-Mamba	3.93	**0.0048**	3.269	3.964	2.982	3.373	4.267	0.761	0.951	2.267	0.771
	SBM	3.93	**0.0048**	3.412	**4.135**	**3.253**	**3.616**	**4.737**	0.820	0.981	3.503	0.891

Table 2: *Results on DNS Real Recordings. Intrusive metrics are not included due to the absence of clean ground truth.*

Test set	Model	SIG↑	BAK↑	OVRL↑	P808 MOS↑	NISQA↑
DNS Real Recordings	SB-NCSN++(50)	3.369	3.939	3.052	3.685	3.604
	SB-NCSN++(10)	3.332	3.933	3.024	3.723	3.865
	SB-NCSN++(1)	3.297	3.924	2.979	3.514	3.541
	SBCTM	3.224	3.841	2.894	3.506	3.647
	SB-UFOGen	3.255	3.838	2.911	3.485	3.196
	ZipEnhancer	3.323	3.918	3.002	3.639	3.264
	Mamba-base	3.284	3.974	2.993	3.683	3.644
	FM-Mamba	3.32	4.031	3.052	3.562	3.797
	SBM	**3.459**	**4.106**	**3.198**	**3.742**	**3.937**

3.2. Datasets and Metrics

The training datasets used in our experiments include clean speeches, noises and room impulse responses (RIRs). Clean speeches are collected from DNS Challenge clean data [26], AI-Shell3 [27] and LibriSpeech [28], totaling around 800 hours. Noises are collected from FSD50K [29], DNS Challenge [26] and self-collected noises from various environments. RIRs are collected from SLR26/28 [30] and simulated using pyroomacoustics [31]. The paired degraded speeches are simulated using the above clean speeches, noises and RIRs with a signal-to-noise ratio (SNR) in $[-10, 20]$ that can cover a wide range of daily-life scenarios. All audio data is pre-processed as 16 kHz.

We evaluate SBM on four benchmark test sets: DNS Challenge Real Recordings, Synthetic With Reverb, Synthetic Without Reverb [26] and VoiceBank-Demand test set [32].

We adopt a set of metrics to evaluate speech enhancement: DNSMOS [33] (*SIG* for signal quality, *BAK* for noise quality, *OVRL* and *P808MOS* for overall quality), *NISQA* [34] for perceptual quality, *SpeechBERTScore* [35] for semantic similarity, *Similarity* [36] for speaker cosine similarity, and traditional alignment-sensitive metrics (*PESQ* [37], *ESTOI* [38]). We use the same pretrained model as [36] for *SpeechBERTScore* and *Similarity*. Although employing a data prediction loss, SBM reconstructs speech details via trajectory guidance rather than yielding statistical averages [1]. Since strictly alignment-sensitive metrics often penalize deviated structures more heavily than over-smoothed predictions [39], alignment-agnostic metrics (*SIG*, *BAK*, *OVRL*, *P808MOS*, *NISQA*, *SpeechBERTScore*, *Similarity*) serve as our primary indicators of model performance, alongside alignment-sensitive metrics for a comprehensive assessment.

3.3. Baseline Models

We validate SBM by comparing it with a group of baselines:

SB-NCSN++ [2]: The classical model of SB for SE. We include three versions for 50-steps inference (**SB-NCSN++(50)**), 10-steps inference (**SB-NCSN++(10)**) and 1-step inference (**SB-NCSN++(1)**). The 1-step version follows the same inference strategy as SBM. Due to the absence of public checkpoints, we re-trained models following the paper and open-sourced codes using the same datasets for SBM. We also tried different losses following the design of SBM and literature [2, 7, 8], and report SB-NCSN++ baselines with the best results.

SBCTM [6]: Built on top of [2, 4], SBCTM involves consistency trajectory modeling [14] to achieve 1-step inference. We used their open-sourced checkpoint in our experiments.

SB-UFOGen [5]: Built on top of [2], SB-UFOGen incorporates adversarial training [13] to achieve 1-step inference. Due to the absence of open-sourced checkpoints, we re-trained this baseline as we did for SB-NCSN++.

ZipEnhancer [20]: A state-of-the-art discriminative model. They published checkpoint for the DNS dataset [40]. For the VoiceBank-Demand test set, lacking a public checkpoint, we computed metrics using the officially released inference results [41].

Mamba-base: To justify the SB training paradigm, this baseline refers to a predictive-mapping trained Mamba-based model. For a fair comparison, we trained this baseline using the same datasets and the same Mamba-based backbone as in SBM.

FM-Mamba: To align with recent generative modeling trends and strengthen our evaluation, we introduce a flow matching (FM) [42] variant. This baseline shares the exact Mamba backbone and one-step inference protocol with SBM, but computes the intermediate states x_t via FM instead of the SB formulation. Specifically, we computed the states x_t using the OT-CFM formulation [43, 44]. This baseline aims to validate whether the SB objective yields superiority over the FM paradigm in our experiments.

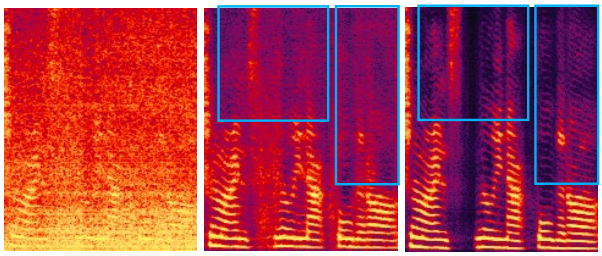


Figure 2: *Spectrogram comparison in a challenging real-world case. SBM successfully reconstructs frequency harmonics, reflecting learned structural priors.*

3.4. Experimental Results and Discussion

As shown in Table 1 and Table 2, SBM demonstrates strong generalization in complex acoustic environments. On the DNS Real Recordings that represent in-the-wild degraded audio, SBM achieves the highest scores across all metrics. Similarly, on the DNS With Reverb, SBM leads on *SIG*, *BAK*, *OVRL*, *P808MOS*, *SpeechBERTScore*, *Similarity*, *PESQ* and *ESTOI*. These results indicate the effectiveness of SBM in jointly addressing real-world noise and reverberation artifacts.

On the DNS No Reverb and VoiceBank-Demand test sets that only include noise artifacts, SBM performs competitively against the discriminative model ZipEnhancer. While the results of ZipEnhancer demonstrate the efficacy of its multi-objective training (e.g., via PESQ-based GAN and phase losses) [20], SBM achieves on-par performance via trajectory-guided reconstruction. Furthermore, by incorporating 25% double-talk scenarios during training to preserve secondary speakers, SBM diverges from test cases requiring single-speaker isolation. Consequently, this discrepancy penalizes SBM on certain metrics like *SpeechBERTScore*, which reflects a strategic design choice for realistic multi-speaker environments rather than a deficit in enhancement capability.

According to Table 1, SBM achieves the lowest real-time factor (RTF). Beyond computational efficiency, it maintains a small algorithmic latency (see Sec 2.2), suggesting its potential for practical low-latency applications.

We further examine the impact of the training paradigm of SBM. Although employing a deterministic one-step inference akin to conventional prediction models [44], SBM receives principled generative guidance by training using the SB OT trajectory. As shown in Figure 2, under severe real-world degradation, SBM successfully reconstructs mid-to-high frequency harmonics, whereas the discriminative baseline exhibits typical over-smoothing. This implies that SBM learns the structural priors, enabling the reconstruction of fine-grained harmonics rather than defaulting to a statistical average.

The empirical results also justify the advantages of the Mamba architecture and the SB training paradigm. SB-NCSN++, SBCTM and SB-UFOGen are all built based on the NCSN++ backbone that is most commonly applied in SB for SE models. While CTM and adversarial training techniques accelerate inference while maintaining model performance [5, 6], they fail in addressing the size and efficiency problems intrinsic to the NCSN++ structure. In comparison, SBM brings significant improvements, showing the efficacy of model structural change under the scope of SB training paradigm. On the other hand, by altering the training paradigm, SBM outperforms Mamba-base and FM-Mamba without RTF degradation. Although the incorporation of timestep embedding slightly enlarges the SBM model, this barely influences inference efficiency. The same backbone structure also implies that edge-side optimization techniques for Mamba are also applicable to SBM, further facilitating the deployment of SBM.

Table 3: *Ablation study results on DNS Real Recordings. (M) stands for the mapping training paradigm and (SB) stands for the SB training paradigm.*

Model	Parameters	SIG↑	BAK↑	OVRL↑	P808 MOS↑	NISQA↑
MHSA*(M)*	8.37M	2.975	3.809	2.676	3.233	2.358
MHSA*(SB)*	8.41M	**3.221**	**3.941**	**2.927**	**3.60**	**3.24**
LSTM*(M)*	17.38M	3.083	3.904	2.799	3.331	2.585
LSTM*(SB)*	17.42M	**3.289**	**3.965**	**3.00**	**3.674**	**3.819**
Mamba*(M)*	3.61M	3.284	3.974	2.993	3.683	3.644
Mamba*(SB)*	3.93M	**3.459**	**4.106**	**3.198**	**3.742**	**3.937**

3.5. Ablation Studies

To further evaluate the synergy between backbones and training paradigms, we conducted ablation studies comparing Mamba against Multi-Head Self-Attention (MHSA) and Long Short-Term Memory (LSTM). We replaced all Mamba blocks in Figure 1 with MHSA or LSTM, and trained models using the SB paradigm and the mapping paradigm respectively. To provide each backbone with a comparable capacity for a fair comparison, the design, hidden dimensions and hyper-parameters of MHSA and LSTM blocks are carefully configured based on their structural correspondence to Mamba [15, 45, 46], leading to the different model sizes shown in Table 3. Table 3 shows the comparison results on DNS Real Recordings.

The results show that the SB paradigm consistently leads across all architectures, demonstrating the effectiveness of training on intermediate states x_t as a form of trajectory-based supervision. Unlike static point-to-point mapping, SB leverages the optimal transport path to provide a continuous-time view of the speech enhancement evolution process. Notably, Mamba achieves superior performance than MHSA and LSTM, indicating that its selective state-space mechanism could be more suitable for internalizing the evolution dynamics. While future work will further investigate the contribution of the trajectory modeling-based training, these results highlight the effective synergy between the SB paradigm and the Mamba architecture.

4. Conclusion

In this paper, we introduced Schrödinger Bridge Mamba (SBM), a novel one-step speech enhancement model guided by the generative Schrödinger Bridge (SB) paradigm and the Mamba architecture. Extensive evaluations demonstrate that SBM delivers state-of-the-art performance in complex real-world and reverberant scenarios, while maintaining a highly competitive real-time factor and a lightweight model size. Furthermore, ablation studies suggest that this robust performance likely benefits from the structural alignment between SB's optimal transport trajectory and Mamba's continuous-time dynamics. Moving forward, we aim to extend SBM to broader audio processing tasks like super-resolution and semantic-level restoration, and further investigate its underlying mechanism, thereby offering valuable insights for achieving high-fidelity speech processing without compromising the efficiency required by real-world applications.

5. Use of Generative AI Disclosure

In compliance with ISCA policy, the authors declare that generative AI tools were utilized solely for editing and polishing the language of this manuscript to improve readability. No generative AI was used to generate core ideas, design experiments, write any major part, or produce data analyses. The authors take full responsibility for the content and integrity of this work.

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